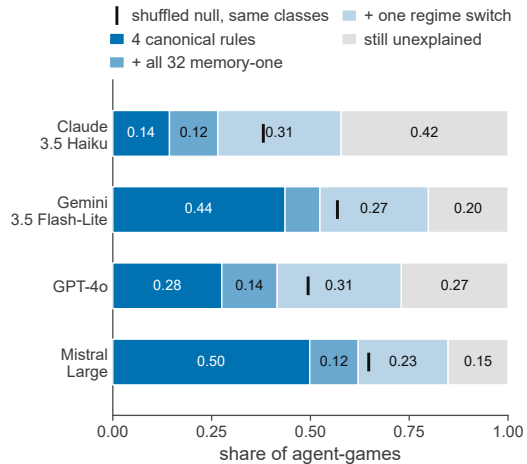
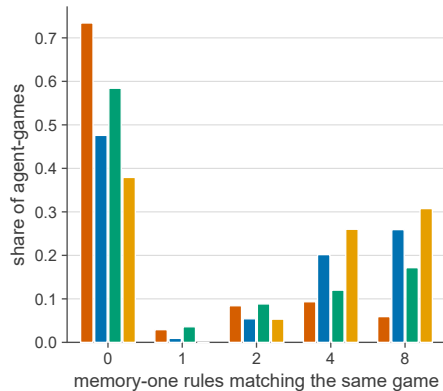


a

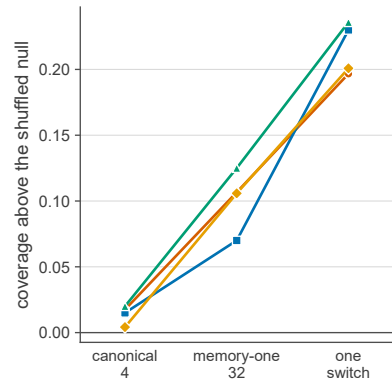
How far a wider vocabulary gets

**b**

The wider vocabulary is under-identified

**c**

Excess over chance



● Claude 3.5 Haiku ■ Gemini 3.5 Flash-Lite ▲ GPT-4o ◆ Mistral Large

The classes in a are nested, so the segments are disjoint and stack to one. The tick is where a shuffled version of the same play already reaches with the same three classes; coverage below that tick is not evidence of structure. Ten rounds rarely visit all four conditioning states, so rules that differ only in an unvisited state are indistinguishable: in b most matched games are matched by 4 or 8 of the 32 rules at once, and only 89 of 4,800 are pinned to exactly one. Panel c is the cumulative excess, and it is what makes the widening worth doing at all.